Question	Answer	Marks	Guidance
4(a)(i)	magnitudes (of electric fields) are equal	B1	**Allow** 'strength' for 'magnitude'
	<u>Any two from:</u> • (electric field) due to proton is to the right • (electric field) due to electron is to the right • directions (of electric fields) are the same	B2	**Allow** towards electron / away from proton **Allow** towards electron / away from proton. **Do not allow** if both directions given to the left. If all three points are attempted, and two are correct and one is incorrect, treat as one contradiction and award 1/2 (CON).
4(a)(ii)	magnitudes (of electric potentials) are equal	B1	
	<u>Any two from:</u> • (electric potential) due to proton is positive • (electric potential) due to electron is negative • <u>signs</u> (of electric potentials) are opposite	B2	**Do not allow** if the signs are the wrong way around. **Do not allow** 'in opposite directions' If all three points are attempted, and two are correct and one is incorrect, treat as one contradiction and award 1/2 (CON). If BP3 is incorrect purely due to use of 'in opposite directions', this does not negate BP1 and BP2.
4(b)(i)	$F = Q_1Q_2 / 4\pi\varepsilon_0R^2$	C1	C1 mark is for any correct general statement of Coulomb's law, using standard symbols. **Allow** Q, q and e for Q_1 and Q_2. **Allow** x or r for R.
	$F = e^2 / 16\pi\varepsilon_0x^2$	A1	Treat correct answer as AFC. **Ignore** sign
4(b)(ii)	$E_P = 2Fx$	A1	**Ignore** sign